

# Movie\_Ratings\_Statistics\_Step\_2

## Importing Data.

Dataset obtained from: <https://www.kaggle.com/datasets/stefanoleone992/rotten-tomatoes-movies-and-critic-reviews-dataset> (Data has been scraped from the publicly available website <https://www.rottentomatoes.com> as of 2020-10-31.)

I have two file formats, .csv and .xlsx. Importing .xlsx (slower to read) to main original data types.

```
# rotten_tomatoes_df <- read.csv("rotten_tomatoes_movies.csv")
rotten_tomatoes_df <- read_excel("rotten_tomatoes_movies.xlsx")

summary(rotten_tomatoes_df)
```

```
## rotten_tomatoes_link movie_title      movie_info      critics_consensus
## Length:17712          Length:17712      Length:17712      Length:17712
## Class :character      Class :character  Class :character  Class :character
## Mode  :character      Mode  :character  Mode  :character  Mode  :character
##
##
##
## content_rating      genres          directors          authors
## Length:17712        Length:17712      Length:17712      Length:17712
## Class :character    Class :character  Class :character  Class :character
## Mode  :character    Mode  :character  Mode  :character  Mode  :character
##
##
##
## actors              original_release_date
## Length:17712        Min.   :1914-06-01 00:00:00.00
## Class :character    1st Qu.:1993-05-08 18:00:00.00
## Mode  :character    Median :2006-10-27 00:00:00.00
##                    Mean   :2000-01-27 06:43:23.07
##                    3rd Qu.:2013-11-01 00:00:00.00
##                    Max.   :2020-09-30 00:00:00.00
##                    NA's    :1166
## streaming_release_date runtime      production_company
## Min.   :1998-07-07 00:00:00.0 Min.   : 5.0      Length:17712
## 1st Qu.:2015-09-29 00:00:00.0 1st Qu.: 90.0      Class :character
## Median :2016-08-10 00:00:00.0 Median : 99.0      Mode  :character
## Mean   :2016-02-06 19:49:46.7 Mean   :102.2
## 3rd Qu.:2017-03-21 00:00:00.0 3rd Qu.:111.0
## Max.   :2020-10-29 00:00:00.0 Max.   :266.0
## NA's    :384                NA's    :314
## tomatometer_status tomatometer_rating tomatometer_count audience_status
```

```
## Length:17712      Min.   : 0.00      Min.   : 5.00      Length:17712
## Class :character  1st Qu.: 38.00      1st Qu.: 12.00      Class :character
## Mode :character   Median : 67.00      Median : 28.00      Mode :character
##                   Mean    : 60.88      Mean    : 57.14
##                   3rd Qu.: 86.00      3rd Qu.: 75.00
##                   Max.    :100.00      Max.    :574.00
##                   NA's    :44         NA's    :44
## audience_rating  audience_count  tomatometer_top_critics_count
## Min.   : 0.00      Min.   : 5         Min.   : 0.00
## 1st Qu.: 45.00      1st Qu.: 708       1st Qu.: 3.00
## Median : 63.00      Median : 4277       Median : 8.00
## Mean    : 60.55      Mean    : 143940     Mean    :14.59
## 3rd Qu.: 78.00      3rd Qu.: 24988     3rd Qu.:23.00
## Max.    :100.00      Max.    :35797635    Max.    :69.00
## NA's    :296        NA's    :297
## tomatometer_fresh_critics_count tomatometer_rotten_critics_count
## Min.   : 0.00      Min.   : 0.0
## 1st Qu.: 6.00      1st Qu.: 3.0
## Median : 16.00      Median : 8.0
## Mean    : 36.37      Mean    : 20.7
## 3rd Qu.: 44.00      3rd Qu.: 24.0
## Max.    :497.00      Max.    :303.0
##
```

## Data Preparation and Cleaning.

```
# Check for missing values.
sum(is.na(rotten_tomatoes_df$tomatometer_rating))
```

```
## [1] 44
```

```
sum(is.na(rotten_tomatoes_df$audience_rating))
```

```
## [1] 296
```

```
sum(is.na(rotten_tomatoes_df$original_release_date))
```

```
## [1] 1166
```

```
# Drop rows with missing critic or audience ratings.
cleaned_rotten_tomatoes_data <- rotten_tomatoes_df %>%
  drop_na(tomatometer_rating, audience_rating, original_release_date)

# Select relevant columns
columns_selected <- c('movie_title', 'content_rating', 'genres', 'directors', 'actors',
  'original_release_date', 'runtime', 'production_company',
  'tomatometer_rating',
  'audience_rating')
cleaned_rotten_tomatoes_data <- cleaned_rotten_tomatoes_data %>% select(all_of(columns_selected))
```

```

# Extract year from release date
cleaned_rotten_tomatoes_data <- cleaned_rotten_tomatoes_data %>%
  mutate(release_year = as.numeric(format(original_release_date, "%Y")))

# Create score discrepancy variable
cleaned_rotten_tomatoes_data <- cleaned_rotten_tomatoes_data %>%
  mutate(score_discrepancy = tomatometer_rating - audience_rating)

```

## Display Cleaned Dataset.

```

# Display structure of cleaned dataset
glimpse(cleaned_rotten_tomatoes_data)

```

```

## Rows: 16,371
## Columns: 12
## $ movie_title      <chr> "Percy Jackson & the Olympians: The Lightning Th~
## $ content_rating   <chr> "PG", "R", "R", "NR", "G", "PG-13", "NR", "NR", ~
## $ genres           <chr> "Action & Adventure, Comedy, Drama, Science Fict~
## $ directors        <chr> "Chris Columbus", "Nicole Holofcener", "Blake Ed~
## $ actors           <chr> "Logan Lerman, Brandon T. Jackson, Alexandra Dad~
## $ original_release_date <dtm> 2010-02-12, 2010-04-30, 1979-10-05, 1957-04-13, ~
## $ runtime          <dbl> 119, 90, 122, 95, 127, 109, 80, 92, 103, 97, 100~
## $ production_company <chr> "20th Century Fox", "Sony Pictures Classics", "W~
## $ tomatometer_rating <dbl> 49, 87, 67, 100, 89, 8, 96, 96, 20, 82, 80, 91, ~
## $ audience_rating  <dbl> 53, 64, 53, 97, 74, 37, 86, 79, 87, 40, 66, 79, ~
## $ release_year     <dbl> 2010, 2010, 1979, 1957, 1954, 2008, 1935, 1957, ~
## $ score_discrepancy <dbl> -4, 23, 14, 3, 15, -29, 10, 17, -67, 42, 14, 12, ~

```

```

# Display summary of cleaned dataset
summary(cleaned_rotten_tomatoes_data)

```

```

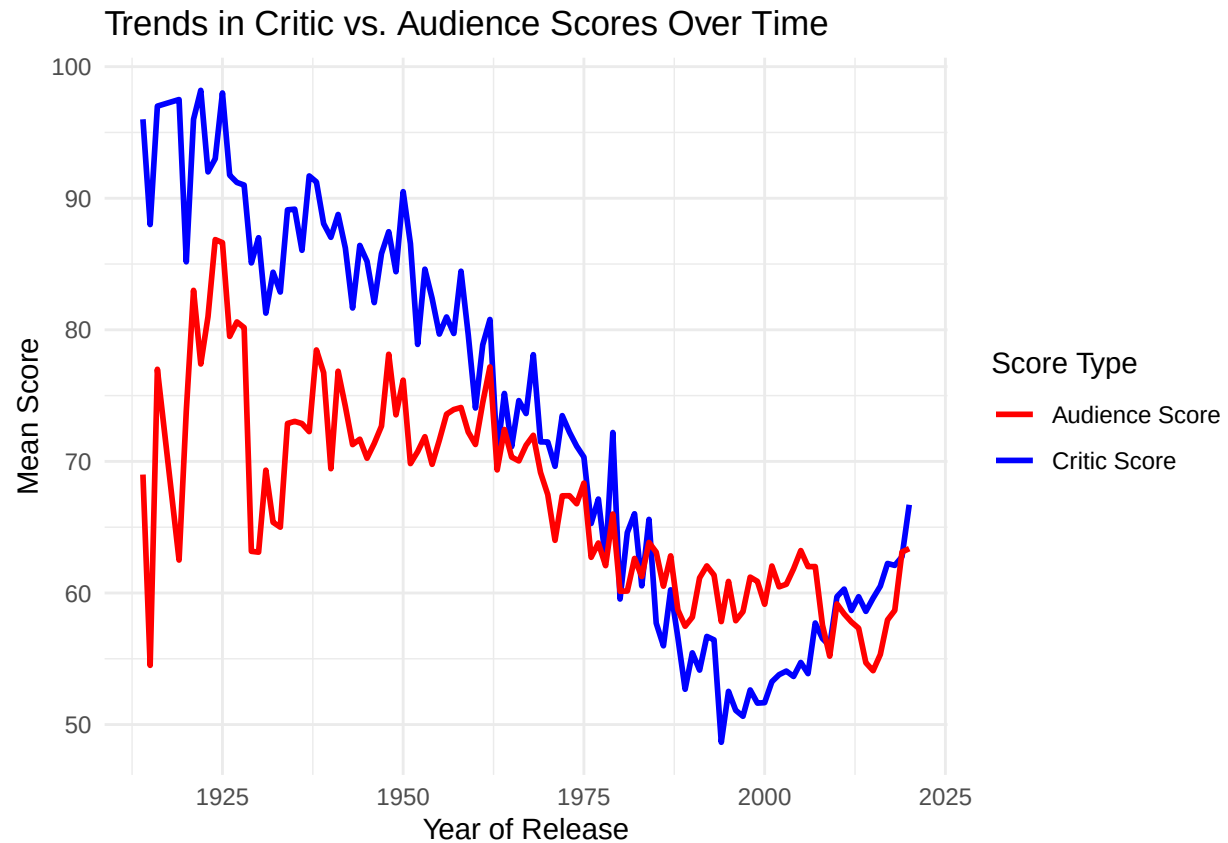
## movie_title      content_rating      genres      directors
## Length:16371     Length:16371     Length:16371 Length:16371
## Class :character  Class :character  Class :character Class :character
## Mode :character   Mode :character   Mode :character   Mode :character
##
##
##
## actors           original_release_date      runtime
## Length:16371     Min. :1914-06-01 00:00:00.000 Min. : 8.0
## Class :character  1st Qu.:1993-02-12 00:00:00.000 1st Qu.: 91.0
## Mode :character   Median :2006-10-06 00:00:00.000 Median :100.0
##                   Mean :1999-12-13 04:38:44.794 Mean :102.8
##                   3rd Qu.:2013-10-11 00:00:00.000 3rd Qu.:111.0
##                   Max. :2020-09-30 00:00:00.000 Max. :266.0
##                   NA's :134
## production_company tomatometer_rating audience_rating release_year
## Length:16371       Min. : 0.00      Min. : 0.00      Min. :1914
## Class :character   1st Qu.: 38.00    1st Qu.: 45.00    1st Qu.:1993

```

```
## Mode :character Median : 66.00 Median : 63.00 Median :2006
## Mean : 60.47 Mean : 60.73 Mean :1999
## 3rd Qu.: 85.00 3rd Qu.: 78.00 3rd Qu.:2013
## Max. :100.00 Max. :100.00 Max. :2020
##
## score_discrepancy
## Min. : -95.000
## 1st Qu.: -13.000
## Median : 2.000
## Mean : -0.262
## 3rd Qu.: 14.000
## Max. : 86.000
##
```

```
# Aggregate the data by year to get mean critic and audience scores.
trend_data <- cleaned_rotten_tomatoes_data %>%
  group_by(release_year) %>%
  summarize(mean_critic_score = mean(tomatometer_rating, na.rm = TRUE),
            mean_audience_score = mean(audience_rating, na.rm = TRUE))

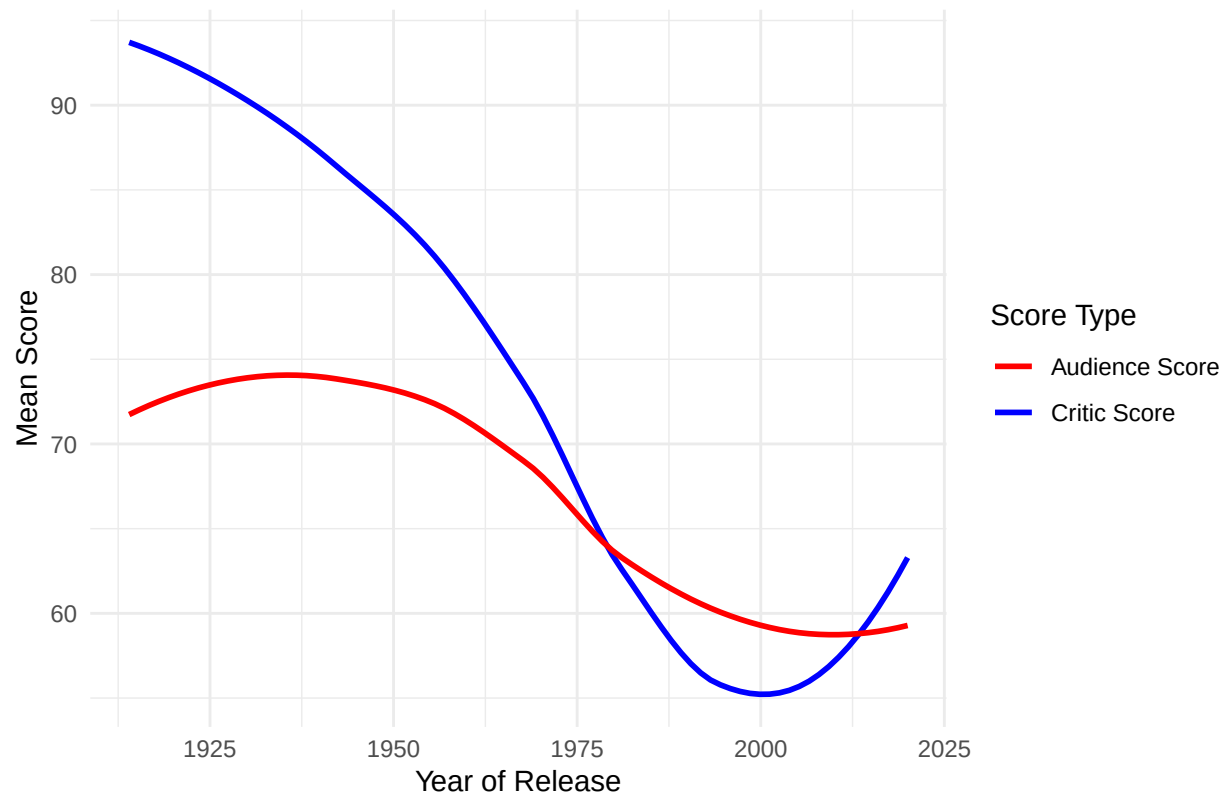
# Plot trends in critic vs. audience scores over time.
ggplot(trend_data, aes(x = release_year)) +
  geom_line(aes(y = mean_critic_score, color = "Critic Score"), linewidth = 1) +
  geom_line(aes(y = mean_audience_score, color = "Audience Score"), linewidth = 1) +
  labs(title = "Trends in Critic vs. Audience Scores Over Time",
       x = "Year of Release",
       y = "Mean Score",
       color = "Score Type") +
  scale_color_manual(values = c("Critic Score" = "blue", "Audience Score" = "red")) +
  theme_minimal()
```



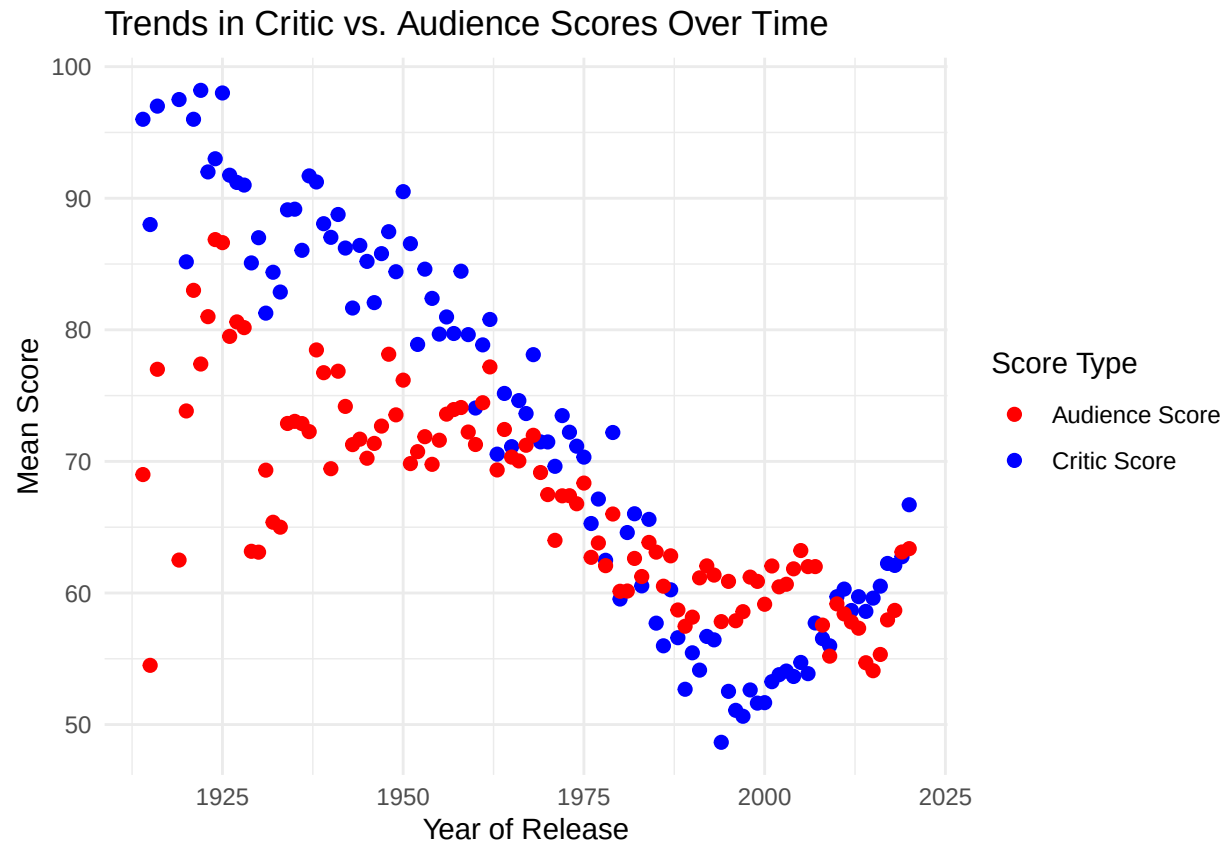
```
# Plot trend lines
ggplot(trend_data, aes(x = release_year)) +
  geom_smooth(aes(y = mean_critic_score, color = "Critic Score"), method = "loess", se = FALSE) +
  geom_smooth(aes(y = mean_audience_score, color = "Audience Score"),
    method = "loess", se = FALSE) +
  labs(title = "Trends in Critic vs. Audience Scores Over Time",
    x = "Year of Release",
    y = "Mean Score",
    color = "Score Type") +
  scale_color_manual(values = c("Critic Score" = "blue", "Audience Score" = "red")) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
```

Trends in Critic vs. Audience Scores Over Time

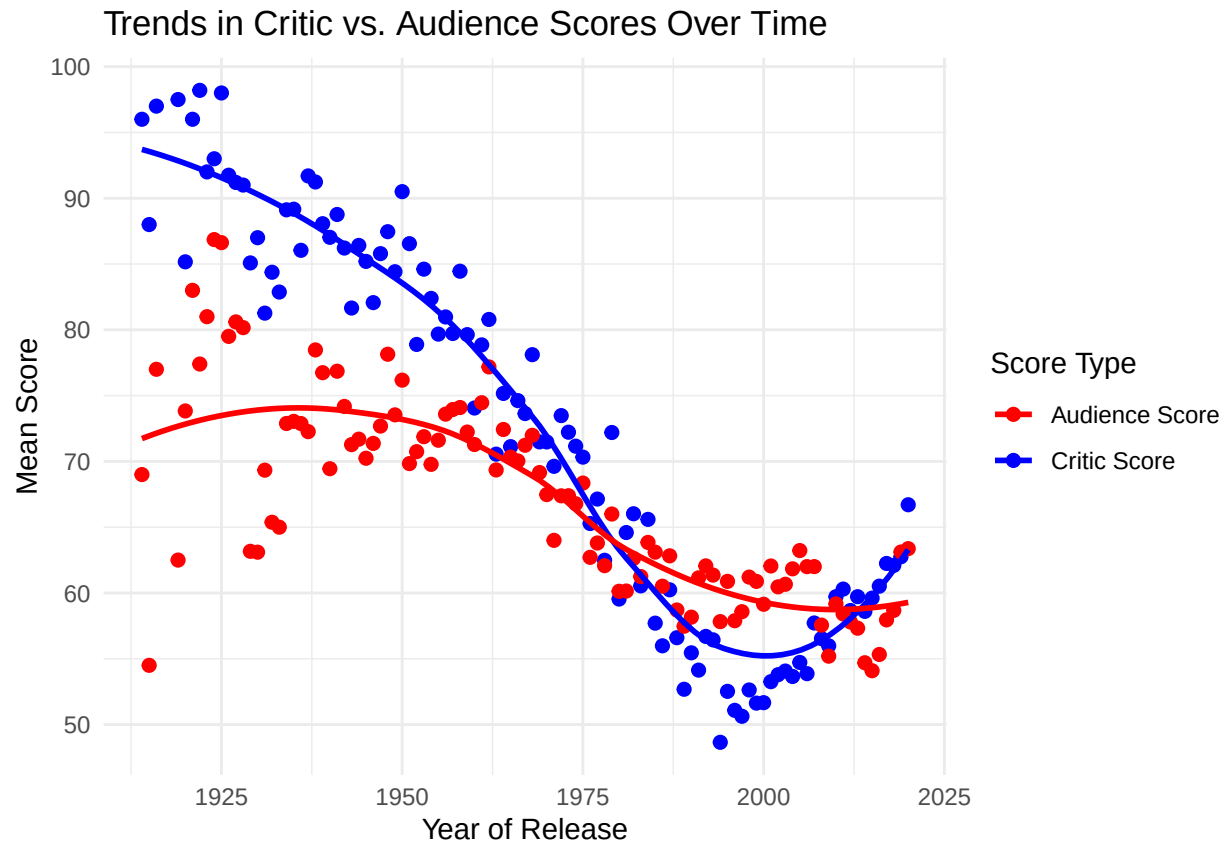


```
# Plot scatter plots.
ggplot(trend_data, aes(x = release_year)) +
  geom_point(aes(y = mean_critic_score, color = "Critic Score"), size = 2) +
  geom_point(aes(y = mean_audience_score, color = "Audience Score"), size = 2) +
  labs(title = "Trends in Critic vs. Audience Scores Over Time", x = "Year of Release",
        y = "Mean Score", color = "Score Type") +
  scale_color_manual(values = c("Critic Score" = "blue", "Audience Score" = "red")) + theme_minimal()
```



```
# Plot scatter plots with trend lines
ggplot(trend_data, aes(x = release_year)) +
  geom_point(aes(y = mean_critic_score, color = "Critic Score"), size = 2) +
  geom_point(aes(y = mean_audience_score, color = "Audience Score"), size = 2) +
  geom_smooth(aes(y = mean_critic_score, color = "Critic Score"), method = "loess", se = FALSE) +
  geom_smooth(aes(y = mean_audience_score, color = "Audience Score"), method = "loess", se = FALSE) +
  labs(title = "Trends in Critic vs. Audience Scores Over Time",
       x = "Year of Release",
       y = "Mean Score",
       color = "Score Type") +
  scale_color_manual(values = c("Critic Score" = "blue", "Audience Score" = "red")) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
```



### What information is not self-evident?

Means of critic and audience scores are grouped by movies' release year.

Rotten Tomatoes website was launched on August 1998.

Critic and audience scores rate movies dating back to 1914.

The specific impact of different genres (a movie can be classified in more than one genre), directors, and actors (more than one) on the score discrepancies.

The influence of budget and box office performance on these scores (No provided in dataset).

### What are different ways you could look at this data?

Analyzing score discrepancies by genre and release year.

Investigating specific directors or actors with consistent score gaps.

Comparing blockbuster movies to independent films.

### How do you plan to slice and dice the data?

Group data by genre, director, and release year.

## **How could you summarize your data to answer key questions?**

Use summary statistics (mean, median, standard deviation). Create pivot tables for aggregating scores by genre, director, etc.

## **What types of plots and tables will help you to illustrate the findings to your questions?**

Bar plots for score discrepancies by genre.

Line plots to show trends over time.

Box plots to compare score distributions between blockbusters and indie films.

## **Do you plan on incorporating any machine learning techniques to answer your research questions? Explain.**

Right now, I am not sure which machine learning techniques I can use. Week 10 will cover machine learning concepts.

## **Questions for future steps.**

How and where can I get movie budget, and box office performance.

How to filter genre, director, actor,

How to arrange data by different categories to reveal patterns.

Reduce data to focus on key variables relevant to the research questions.

Create new variables to measure and compare differences.

Summary statistics highlighting trends and differences across categories.

Using pipe operator to streamline data manipulation and transformation steps.